

Series and infinite products – 7

Infinite products

Homework

12 сентября 2026 года

Theoretical minimum. A product $\prod(1 + a_n)$ converges when the partial products tend to a non-zero limit; for $a_n > -1$ this is equivalent to convergence of $\sum \ln(1 + a_n)$, and absolute convergence is equivalent to $\sum |a_n| < \infty$. Because $\ln(1 + a) = a - \frac{a^2}{2} + O(a^3)$, the pair of series $\sum a_n$ and $\sum a_n^2$ decides everything, and a product may diverge to zero while $\sum a_n$ converges. A rational factor telescopes whenever its numerator and denominator have roots differing by integers, and the answer is a ratio of gamma values; a factor of the form $1 - \frac{1}{a_k}$ with $a_{k+1} = a_k^2 - 2$ telescopes after $a_k = z_k + z_k^{-1}$. The classical products are Euler's for the sine, Wallis' and Viète's; over primes, a product is the Euler product of a Dirichlet series.

1. (Gazeta Matematica, problem 529) Prove that

$$\prod_{n=2}^{\infty} e^{n^2 + \frac{1}{2}} \left(1 - \frac{1}{n^2}\right)^{n^4} = \pi \exp\left(-\frac{7}{4} + \frac{3\zeta(3)}{\pi^2}\right),$$

where ζ denotes the Riemann zeta function.

2. (Putnam 2011) Evaluate $\lim_{x \rightarrow 1^-} \prod_{n=0}^{\infty} \left(\frac{1+x^{n+1}}{1+x^n}\right)^{x^n}$.
3. (Putnam 2015) Compute $\log_2 \left(\prod_{a=1}^{2015} \prod_{b=1}^{2015} (1 + e^{2\pi i ab/2015}) \right)$, where i is the imaginary unit.
4. (Putnam 2021) Let $F_0, F_1, \dots$ be the Fibonacci numbers, with $F_0 = 0$, $F_1 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$. For $m > 2$, let R_m be the remainder when the product $\prod_{k=1}^{F_m-1} k^k$ is divided by F_m . Prove that R_m is also a Fibonacci number.

5. Give an elementary proof of Euler's product for the sine, starting from the factorisation of $\sin nx$ as a polynomial in $\sin x$ and passing to the limit, and deduce $\zeta(2) = \frac{\pi^2}{6}$ by comparing the coefficients of x^2 . (The complex-analytic proof and the derivation of Wallis' formula from it are in Complex analysis – 6.)
6. Evaluate $\prod_{n=1}^{\infty} \left(1 + \frac{1}{n(n+2)}\right)$ and $\prod_{n=2}^{\infty} \frac{n^2-1}{n^2+1}$, and explain in terms of gamma values why the first is rational and the second is not.
7. Prove that $\prod_{n \geq 0} (1 + x^{2^n}) = \frac{1}{1-x}$ for $|x| < 1$, and deduce the binary expansion statement behind it; then compute $\prod_{n \geq 0} (1 + \frac{1}{2^{2^n}})$.
8. Prove Euler's pentagonal-number identity in the form $\prod_{n \geq 1} (1 - x^n) = \sum_{k \in \mathbb{Z}} (-1)^k x^{k(3k-1)/2}$ for $|x| < 1$, or at least verify it up to x^{12} and explain the combinatorial meaning of both sides.
9. Let $p_1 < p_2 < \dots$ be the primes. Prove that $\prod (1 - \frac{1}{p_k}) = 0$ and that $\prod (1 - \frac{1}{p_k^2}) = \frac{6}{\pi^2}$, and say which theorem of the class sheet distinguishes the two cases.
10. Let $a_n > 0$. Prove that $\prod (1 + a_n)$ and $\prod \frac{1}{1-a_n}$ converge or diverge together, and that in the divergent case the partial products of the first grow at least as fast as $\sum a_n$. Give an example with $\sum a_n = \infty$ where $\prod (1 - a_n)$ tends to zero arbitrarily slowly.